Lesson 1: How to optimize your sleep quality

Sleep is important. Yeah, you think you know this, but I suspect you don't realize just how important sleep really is. I'm not even going to go into how you feel, how your brain functions or any of that. Let's talk about your physique. In a randomized cross-over trial by Nedeltcheva et al. (2010), sleeping ~5 hours compared to ~7.5 hours a day in the researchers' lab decreased the proportion of weight lost as fat by 55% and increased the loss of fat-free mass by 60%(!) Sleep deprivation decreased fat loss from 1.4 kg (3.1 lb) to 0.6 kg (1.3 lb) and increased fat-free mass loss from 1.5 kg (3.3 lb) to 2.4 kg (5.3 lb). A study by Wang et al. (2018) found even worse effects of sleep deprivation. Sleeping 40 fewer minutes during the midweek caused the ratio of lean to fat mass loss to shift from a mere ~20% lean mass loss to ~80% lean mass loss. (Technical note: the ratios reported in the paper are incongruent with the absolute values. The authors told us it's due to the difference between the average ratio and the ratio of the averages, but without the full data set, we haven't been able to replicate their calculations.) Last but not least, Jåbekk et al. (2020) compared body composition changes in a group strength training twice a week without changing their sleep practices compared to a group training and also optimizing their sleep quality.

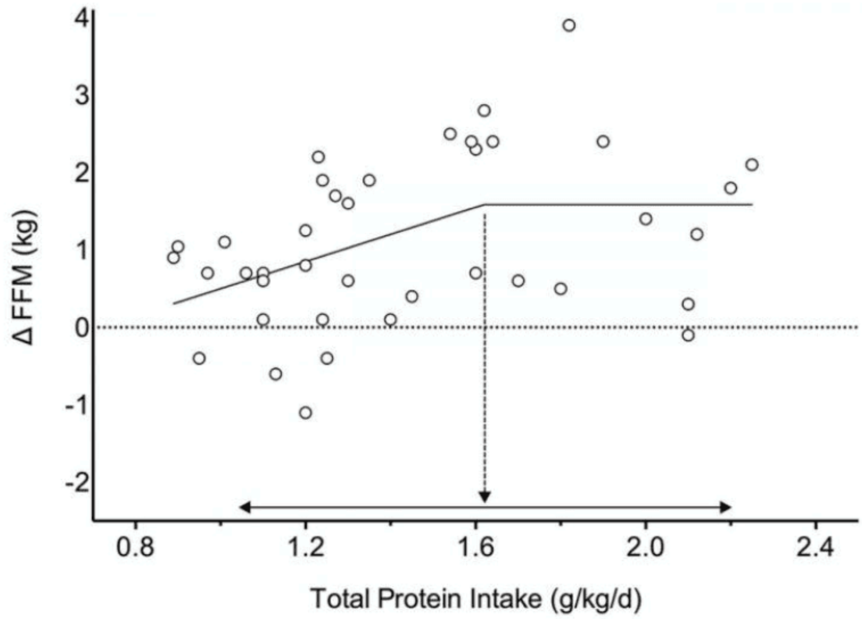
Over 10 weeks, the sleep-optimized group lost 1.8 kg (4 lb) fat compared to 0.8 kg (1.8 lb) gain in the regular training group. Despite being in energy deficit rather than surplus, the sleep optimizers also gained 1.7 kg (3.7 lb) lean body mass instead of 1.3 kg (2.9 lb).

The difference in muscle growth didn't reach formal statistical significance though.

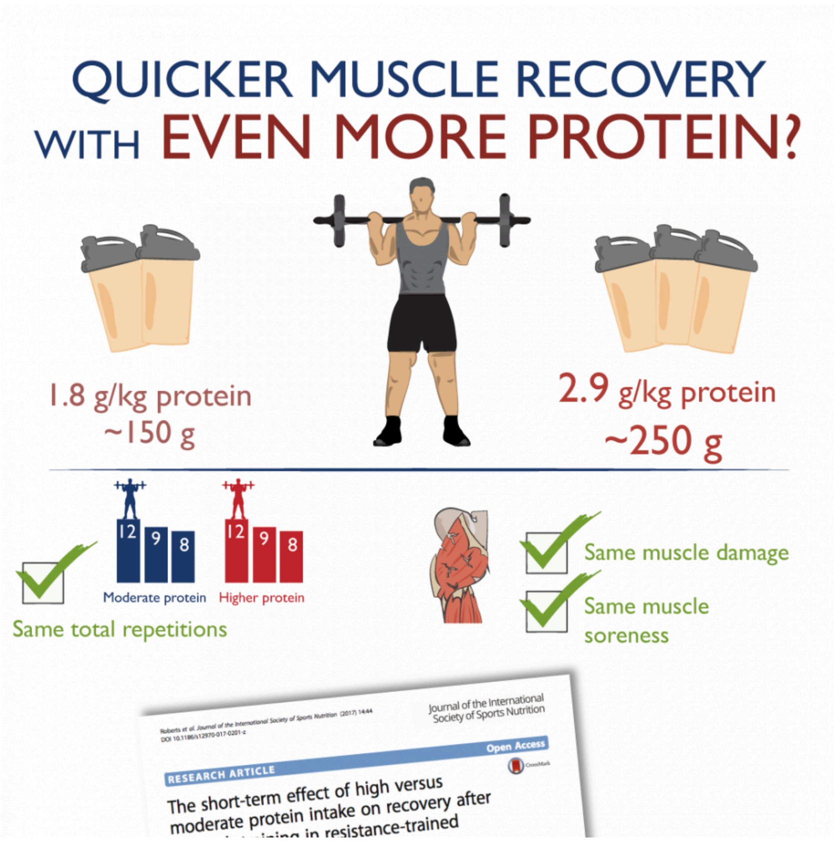
All 3 studies show that even relatively minor sleep deprivation can significantly impair our gains. As such, fixing your sleep can over the course of a year make you gain pounds of muscle and lose pounds of fat without exercise or dieting. So sleep quality is not just important: it's paramount. Both the total duration and the quality of your sleep are arguably much more important than the finetuning of your macros or which variation of an exercise you do. To help you sleep soundly, here are 8 of the tips from my online PT Course's guide on sleep optimization. The full guide contains 14 evidence-based tips.

Lesson 2:How much protein do you need to maximize muscle growth?

Protein is awesome. The word literally means 'of first importance', from Greek. There's a strong research consensus that all strength trainees should be on a high-protein diet. However, more is not always better. So how much protein do we need? In one of my first articles, I compiled all the scientific research on protein intake. Unlike the consensus at the time that you need at the very least one gram protein per pound of total bodyweight, research did not support such high recommendations. All controlled long-term studies and estimates of protein requirements based on nitrogen balance (the gold standard at the time) showed that there were no significant benefits for muscle growth, strength development, fat loss or muscle protein synthesis of going over 1.6 g/kg (0.72 g/lb) per day. I added a statistical margin of safety and recommended a daily protein target of at least 1.8 g/kg per day for the average lifter. That article became highly influential and ranked #1 on Google for 'optimal protein intake' for many years, but it remained controversial. In 2017 I co-authored a meta-analysis of the scientific literature with some of the world's leading fitness researchers. We found a cut-off point at exactly 1.6 g/kg (0.72 g/lb) per day beyond which no further benefits for fat-free mass gains were seen: see the results below.



However, the breakpoint was not quite statistically significant ($p = 0.08$) and the 95% confidence interval spanned to 2.2 g/kg (1 g/lb) per day, so it's possible that protein intakes up to that point confer some small benefits for certain lifters. That same year, my research team also performed a scientific study in collaboration with the University of Cambridge to see if higher protein intakes than 1.8 g/kg per day benefit recovery in the days after a hard workout. They didn't: see the infographic below.



As of January 2025, there's still not a single well-controlled, long-term study finding significant benefits of higher protein intakes than 1.8 g/kg (0.8 g/lb) per day for muscle growth or strength development. However, a few poorly controlled studies and meta-analyses of trends in the uncontrolled data, as well as some data from newer Indicator Amino Acid Oxidation methods support potentially higher intakes.

Most lifters are probably covered by a protein intake of 1.8 to 2.2 g/kg (0.8 to 1.0 g/lb) per day, but certain lifters may benefit from more, in particular vegans, concurrent athletes and trainees on anabolic steroids.